Phrase Mining

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1 Methods

To analyze the terminologies used in the collected corpus and understand the trend of studies, a structured phrase mining approach was applied. This involved a series of natural language processing (NLP) pre-processing steps aimed at improving text uniformity, reducing noise, and improving key-phrase extraction.

Following the project guidelines, only the abstract of the papers categorized as '1', meaning papers related to animal QTL, were considered. The pre-processing steps consisted of text normalization (lowercasing, punctuation removal, and white space stripping), tokenization, stopword removal, lemmatization, and key-phrase extraction. These steps ensure more relevant TF-IDF scores and Word2Vec embeddings, which are the requirements for this project. In addition, they also contribute to the development of automated and expedited curation tools in general.

1.1 Lowercase Text

The purpose of converting all text to lowercase is to ensure uniformity. This step reduces the vocabulary size by preventing the model from treating the same word differently based on capitalization (e.g., "Gene" vs. "gene"). This transformation was applied using Python's built-in lower() function within a lambda function.

1.2 Punctuations Removal

Punctuation marks do not provide semantic meaning and can introduce noise in TF-IDF scoring and Word2Vec embeddings. This step helps to standardize text and improves word co-occurrence patterns, which helps the Word2Vec training. Punctuation was removed using Python's str.translate() function with string.punctuation within a lambda function.

1.3 Tokenization

Tokenization was performed using NLTK's 'word_tokenize' function, which efficiently splits text into individual words (well-known as tokens) while handling punctuation and special characters appropriately. This method ensures that tokens are correctly separated for TF-IDF scoring and Word2Vec training.

1.4 Stop Words Removal

Stopwords were removed using NLTK's predefined stopword list, which filters out common words that do not contribute meaningful information. This step reduces noise in TF-IDF scores by preventing high-frequency, low-value words from dominating the rankings and improves the Word2Vec embeddings by focusing on domain-specific terms.

1.5 Llemmatization

Lemmatization was performed using NLTK's WordNetLemmatizer, which reduces words to their base form while preserving their meaning. This step helps TF-IDF scoring by reducing word variations (e.g., "genes" to "gene") and improves Word2Vec embeddings by ensuring consistent word representations.

1.6 Key-Phrase Extraction

Key-phrases were extracted using Gensim's Phrases model, which identifies frequent multi-word expressions based on co-occurrence statistics. This method detects bigrams and trigrams by analyzing the likelihood of words appearing together compared to their individual occurrences. The model assigns a score to each word pair, and phrases exceeding a defined threshold are merged into a single token, which helps to improve Word2Vec embeddings by capturing phrase-level semantics.

2 Main result

The following sections present key observations from these analyses, highlighting the impact of TF-IDF based term weighting, word similarity relationships using Word2Vec, and extracted key-phrase coverage relative to the trait dictionary.

2.1 Word Cloud Comparison

Word Cloud Analysis Without Key-Phrases

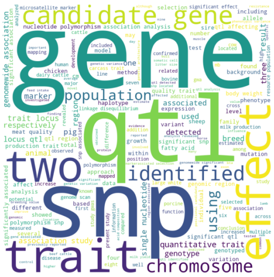

Figure 1: Count-Based Word Cloud

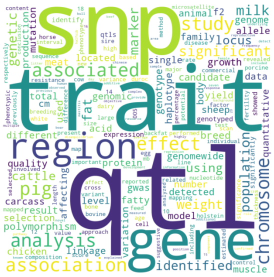

Figure 2: TF-IDF Word Cloud

Comparing the count based and TF-IDF based word clouds without key-phrases, the results align with expectations. The count based approach highlights generic high-frequency terms such as gene, qtl, and trait, making it difficult to extract meaningful trends.

In contrast, TF-IDF weighting highlights more informative but less frequent words such as genomewide and association, indicating its effectiveness in highlighting domain-relevant terms.

Word Cloud With Key-Phrases

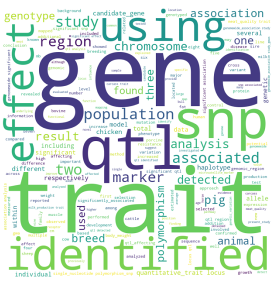

Figure 3: Count-Based Word Cloud

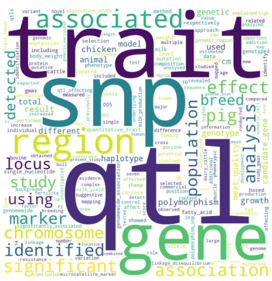

Figure 4: TF-IDF Word Cloud

Applying key-phrase extraction improved word cloud quality. The count based word cloud now keeps multi-word expressions like quantitative_trait and candidate_gene, which makes the terminology more interpretable.

When combined with TF-IDF, high-frequency but less meaningful words lose weight, while domain-specific phrases are highlighted, resulting in a more insightful representation.

These findings confirm my hypothesis: TF-IDF improves term weighting, but key-phrase extraction is essential to retain meaningful multi-word expressions. Without key-phrases, terminology remains fragmented, which limits the interpretability of results.

The best results are obtained when TF-IDF and key-phrases are combined, allowing for more accurate terminology extraction and trend analysis in the dataset.

2.2 Word2Vec Embeddings

The Word2Vec results mostly align with my expectations, showing that related terms appear together, but some unexpected associations were also found.

Without prior biological knowledge, I expected terms like qtl to be grouped with chromosome, mapped, and genomewide, which is consistent with what the model produced. Similarly, trait being associated with growth, carcass, and meat_quality makes sense within the dataset and are positive examples in my opinion.

However, some results were surprising. For example, association is linked to numbers like three, five, and seven, and effect appears near statistical values like 001 and 0001. This suggests that the model is learning frequent co-occurrences rather than true contextual meaning, which could be improved by removing numbers during preprocessing. However, there are words and terminologies that include numbers in them, which is why the numbers were kept.

Overall, the results confirm that Word2Vec captures meaningful relationships, but some noise remains. Better preprocessing rules where numbers that are not part of words and terminologies get removed could potentially help refine embeddings and improve term relevance.

2.3 Trait Dictionary Matching

The key-phrase extractor identified the equivalent of 70.47% of the number of good phrases from the trait dictionary, confirming that the number of good terms were captured. However, the low exact match rate of 1.76% was much lower than expected, suggesting that many extracted phrases differ in structure or wording from those in the trait dictionary.

This indicates that while the method effectively identifies key phrases, variations in phrasing and dataset constraints impact exact matching. Incorporating techniques like fuzzy matching or cosine similarity could improve alignment with the trait dictionary.

References

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Assistance from AI (ChatGPT) was used for refining the structure and wording of this report.